Advanced Database Management Systems (ADBMS)

Assignment 0 & Assignment 1: Histogram Analysis Report

IIIT Bangalore

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1 Advanced Database Management Systems (ADBMS)

1.1 Assignment 0 & Assignment 1: Histogram Analysis Report

2 Assignment 0

2.1 Setup

- **CPU Model & Specification**: 13th Gen Intel(R) Core(TM) i3-1315U (6 cores, 8 threads)
- **RAM Capacity**: 14 GiB
- **HDD / Storage Capacity**: 456 GiB NVMe SSD
- **Operating System**: Ubuntu 26.04 LTS
- **PostgreSQL Engine Version**: `PostgreSQL 18.6 (Ubuntu 18.6-0ubuntu0.26.04.1) on x86_64-pc-lin compiled by gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0, 64-bit`

2.2 Database

- **Database Name**: Join Order Benchmark (2013 Snapshot)
- **Total Tables**: 21
- **Total Relational Tuples**: 74,190,187

2.2.1 Table Cardinalities & Primary Keys

Table Name	Primary Key Column	Cardinality (`count(*)`)
aka_name	id	901,343
aka_title	id	361,472
cast_info	id	36,244,344
char_name	id	3,140,339
comp_cast_type	id	4
company_name	id	234,997
company_type	id	4
complete_cast	id	135,086
info_type	id	113
keyword	id	134,170
kind_type	id	7
link_type	id	18
movie_companies	id	2,609,129
movie_info	id	14,835,720
movie_info_idx	id	1,380,035
movie_keyword	id	4,523,930
movie_link	id	29,997
name	id	4,167,491
person_info	id	2,963,664
role_type	id	12
title	id	2,528,312

3 Assignment 1: Histogram

3.1 Q1. Histogram Analysis for `id` and `title` Columns of `title` Table

3.1.1 1. Corresponding SQL Query

to see what all colums needed form 'pg_stats'

```
\d pg_stats;
```

to see histogram boundary values and other relevent stored in `pg_stats`:

```
SELECT
    attname,
    null_frac,
    n_distinct,
    most_common_vals,
    most_common_freqs,
    histogram_bounds
FROM pg_stats
WHERE tablename = 'title'
  AND attname IN ('id', 'title');
```

3.1.2 2. Kind of Histogram

PSQL uses an Equi Depth Histogram combined with a Most Common Values list. In an equidepth histogram, all buckets contain approximately the same number of rows ($approxN/B$), while bucket boundaries vary in width according to data distribution.

3.1.3 3. Number of Buckets

- Column `id`: **100 buckets** (101 bucket boundary values)
- Column `title`: **100 buckets** (101 bucket boundary values)
- extra - to see the defualt

  ```
  SHOW default_statistics_target;
  ```

3.1.4 4. How Textual Attribute `title` is Put into Histogram

Textual attributes in PostgreSQL are converted into histograms by:

1. sorting the text strings in lexicographical order like bannana and apple will be apple, bannana after sorting

2. Sampling string values from the table like table title has around 2.5M rows , it will only sample around 30K

3. Encoding initial string prefix bytes into floating-point numbers in $[0.0, 1.0]$ range for quantitative interpolation like laptop will be broken to lap with l = 108, a=97, p=112 then by 256 as between 0 and 1 , lap will be around 0.42 , we can then see where a key belongs between two keeys

4. Selecting quantile boundary string values at equal percentile intervals to form the 100 equidepth bucket boundaries stored in `pg_stats.histogram_bounds`.

3.1.5 5. Exact Bucket Boundaries List (From pg_stats)

Bucket Index	Boundary Value (id)	Bucket Interval
B000 (Start)	41	Start of table data
B001	25598	Bucket 1: [41 - 25598]
B002	50067	Bucket 2: [25598 - 50067]
B003	75257	Bucket 3: [50067 - 75257]
B004	101946	Bucket 4: [75257 - 101946]
B005	129223	Bucket 5: [101946 - 129223]
B006	155973	Bucket 6: [129223 - 155973]
B007	181990	Bucket 7: [155973 - 181990]
B008	208153	Bucket 8: [181990 - 208153]
B009	234525	Bucket 9: [208153 - 234525]
B010	259507	Bucket 10: [234525 - 259507]
B011	287845	Bucket 11: [259507 - 287845]
B012	314618	Bucket 12: [287845 - 314618]
B013	340479	Bucket 13: [314618 - 340479]
B014	366364	Bucket 14: [340479 - 366364]
B015	393076	Bucket 15: [366364 - 393076]
B016	418929	Bucket 16: [393076 - 418929]
B017	443204	Bucket 17: [418929 - 443204]
B018	472539	Bucket 18: [443204 - 472539]
B019	497095	Bucket 19: [472539 - 497095]
B020	521015	Bucket 20: [497095 - 521015]
B021	547008	Bucket 21: [521015 - 547008]
B022	571914	Bucket 22: [547008 - 571914]
B023	600124	Bucket 23: [571914 - 600124]
B024	624290	Bucket 24: [600124 - 624290]
B025	648346	Bucket 25: [624290 - 648346]
B026	673077	Bucket 26: [648346 - 673077]
B027	698192	Bucket 27: [673077 - 698192]
B028	723945	Bucket 28: [698192 - 723945]
B029	747971	Bucket 29: [723945 - 747971]
B030	774715	Bucket 30: [747971 - 774715]
B031	801886	Bucket 31: [774715 - 801886]
B032	825206	Bucket 32: [801886 - 825206]
B033	852044	Bucket 33: [825206 - 852044]
B034	878630	Bucket 34: [852044 - 878630]
B035	903227	Bucket 35: [878630 - 903227]
B036	929542	Bucket 36: [903227 - 929542]
B037	951992	Bucket 37: [929542 - 951992]
B038	976177	Bucket 38: [951992 - 976177]
B039	1002223	Bucket 39: [976177 - 1002223]
B040	1027496	Bucket 40: [1002223 - 1027496]
B041	1053013	Bucket 41: [1027496 - 1053013]
B042	1076669	Bucket 42: [1053013 - 1076669]
B043	1101644	Bucket 43: [1076669 - 1101644]
B044	1127312	Bucket 44: [1101644 - 1127312]
B045	1153408	Bucket 45: [1127312 - 1153408]
B046	1177087	Bucket 46: [1153408 - 1177087]
B047	1200809	Bucket 47: [1177087 - 1200809]
B048	1228132	Bucket 48: [1200809 - 1228132]
B049	1251987	Bucket 49: [1228132 - 1251987]
B050	1276194	Bucket 50: [1251987 - 1276194]
B051	1300208	Bucket 51: [1276194 - 1300208]
B052	1325279	Bucket 52: [1300208 - 1325279]
B053	1350079	Bucket 53: [1325279 - 1350079]
B054	1375217	Bucket 54: [1350079 - 1375217]
B055	1399305	Bucket 55: [1375217 - 1399305]
B056	1424915	Bucket 56: [1399305 - 1424915]
B057	1447763	Bucket 57: [1424915 - 1447763]
B058	1472187	Bucket 58: [1447763 - 1472187]
B059	1496066	Bucket 59: [1472187 - 1496066]
B060	1520933	Bucket 60: [1496066 - 1520933]

A. Column `id` Bucket Boundaries (101 values, 100 buckets)

Bucket Index	Boundary Value (`title`)	Lexicographical Interval
B000 (Start)	`...`	Start of text data
B001	`(#1.1302)`	Bucket 1: [`... - (#1.1302)`]
B002	`(#1.219)`	Bucket 2: [`(#1.1302) - (#1.219)`]
B003	`(#1.3662)`	Bucket 3: [`(#1.219) - (#1.3662)`]
B004	`(#1.59)`	Bucket 4: [`(#1.3662) - (#1.59)`]
B005	`(#1.846)`	Bucket 5: [`(#1.59) - (#1.846)`]
B006	`(1968-02-14)`	Bucket 6: [`(#1.846) - (1968-02-14)`]
B007	`(1982-12-14)`	Bucket 7: [`(1968-02-14) - (1982-12-14)`]
B008	`(1994-07-07)`	Bucket 8: [`(1982-12-14) - (1994-07-07)`]
B009	`(1998-03-16)`	Bucket 9: [`(1994-07-07) - (1998-03-16)`]
B010	`(2000-01-22)`	Bucket 10: [`(1998-03-16) - (2000-01-22)`]
B011	`(2001-11-06)`	Bucket 11: [`(2000-01-22) - (2001-11-06)`]
B012	`(2003-08-29)`	Bucket 12: [`(2001-11-06) - (2003-08-29)`]
B013	`(2005-01-19)`	Bucket 13: [`(2003-08-29) - (2005-01-19)`]
B014	`(2006-05-17)`	Bucket 14: [`(2005-01-19) - (2006-05-17)`]
B015	`(2007-07-05)`	Bucket 15: [`(2006-05-17) - (2007-07-05)`]
B016	`(2009-02-02)`	Bucket 16: [`(2007-07-05) - (2009-02-02)`]
B017	`(2010-09-07)`	Bucket 17: [`(2009-02-02) - (2010-09-07)`]
B018	`(2011-08-29)`	Bucket 18: [`(2010-09-07) - (2011-08-29)`]
B019	`(2012-07-21)`	Bucket 19: [`(2011-08-29) - (2012-07-21)`]
B020	`(#20.140)`	Bucket 20: [`(2012-07-21) - (#20.140)`]
B021	`(#2.92)`	Bucket 21: [`(#20.140) - (#2.92)`]
B022	`(#4.172)`	Bucket 22: [`(#2.92) - (#4.172)`]
B023	`(#6.163)`	Bucket 23: [`(#4.172) - (#6.163)`]
B024	`Aamu joutuu pesemään tekojensa jälkipyykkiä`	Bucket 24: [`(#6.163) - Aamu joutuu pesemään tekojensa jälkipyykkiä`]
B025	`Aflevering 2559`	Bucket 25: [`Aamu joutuu pesemään tekojensa jälkipyykkiä - Aflevering 2559`]
B026	`A Little Trip`	Bucket 26: [`Aflevering 2559 - A Little Trip`]
B027	`An Atheist Named Faith`	Bucket 27: [`A Little Trip - An Atheist Named Faith`]
B028	`Arabeskebi Pirosmanis temaze`	Bucket 28: [`An Atheist Named Faith - Arabeskebi Pirosmanis temaze`]
B029	`Attenborough's Ark: Natural World Special`	Bucket 29: [`Arabeskebi Pirosmanis temaze - Attenborough's Ark: Natural World Special`]
B030	`Baldur Blauzahn`	Bucket 30: [`Attenborough's Ark: Natural World Special - Baldur Blauzahn`]
B031	`Best in Show Cakes`	Bucket 31: [`Baldur Blauzahn - Best in Show Cakes`]
B032	`Blues`	Bucket 32: [`Best in Show Cakes - Blues`]
B033	`Brown Before Green`	Bucket 33: [`Blues - Brown Before Green`]
B034	`Captive Audience`	Bucket 34: [`Brown Before Green - Captive Audience`]
B035	`Cheryl Cole's Night In`	Bucket 35: [`Captive Audience - Cheryl Cole's Night In`]
B036	`Cobra Starship`	Bucket 36: [`Cheryl Cole's Night In - Cobra Starship`]
B037	`CPM 22 e Babado Novo`	Bucket 37: [`Cobra Starship - CPM 22 e Babado Novo`]
B038	`Dark Moon`	Bucket 38: [`CPM 22 e Babado Novo -`

B. Column `title` Bucket Boundaries (101 values, 100 buckets)

3.1.6 6. Histogram Visualization

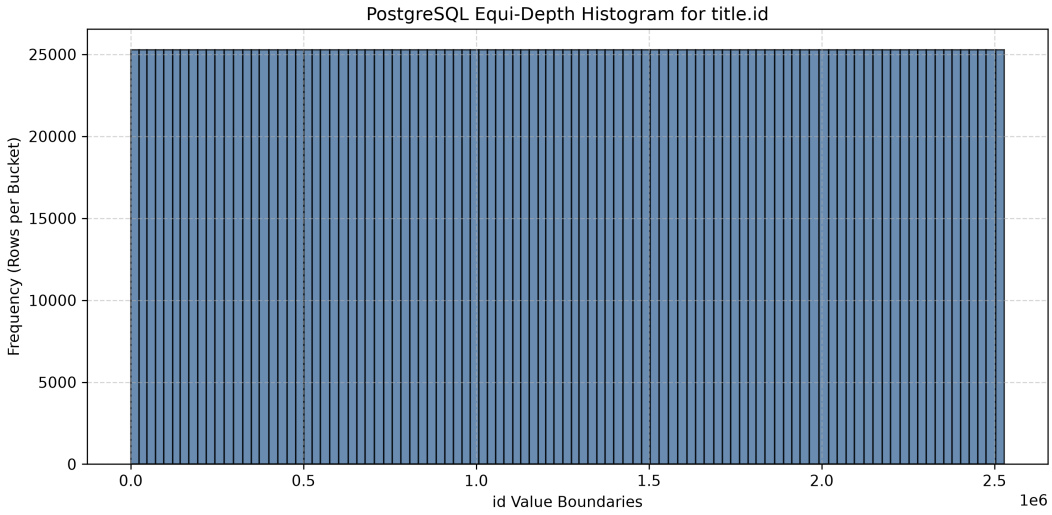


Figure 1: Histogram for title.id

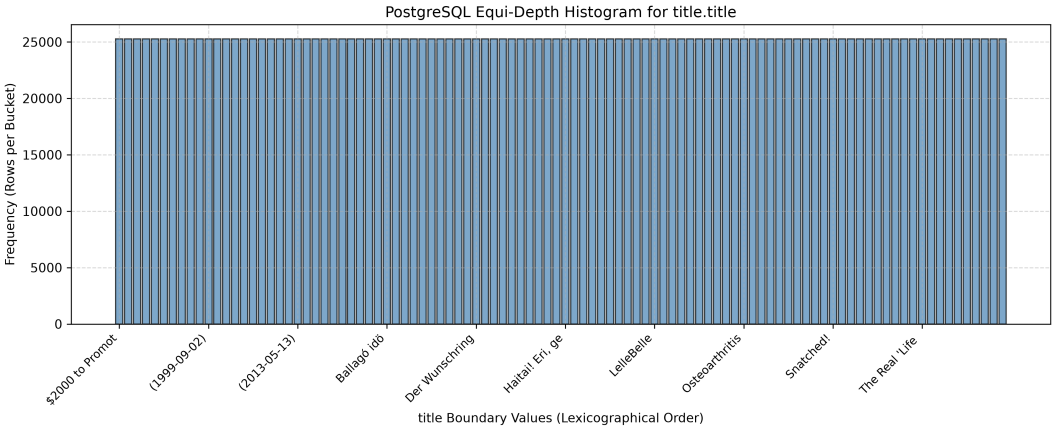


Figure 2: Histogram for title.title

3.1.7 7. Max Selectivity Error

- **Max Selectivity Error**: Since `default_statistics_target` is 100, there are 100 buckets. Therefore, the maximum selectivity error for a range query is bounded by:

$$\text{Max Error} \leq \frac{1}{B} = \frac{1}{100} = 0.01 \quad (1.0\%)$$

For evaluated queries(q9), the observed maximum selectivity error is **0.8364%** (0.008364), which strictly satisfies Max Error $\leq 1.0\%$.

3.1.8 8: Selectivity Estimation for Text Columns

Selectivity estimate is done for range query on text column such as "title" in PSQL such as:

- **Bucket Identification:** The engine scans the `histogram_bounds` array stored under pg_stats to locate the specific bucket where the target string (`'Race'`) is

- **Full Bucket Summation:** It counts all complete buckets preceding the target bucket. Each full bucket represents a fixed fraction of total rows (1/B).
- **Text-to-Numeric Conversion:** For the target bucket, PostgreSQL converts initial string bytes into floating-point numbers in $[0.0, 1.0]$ range for interpolation as explain in part 4
- **Linear Interpolation:** Assuming uniform distribution inside the target bucket, it calculates the target's position using:

$$\text{Fraction} = \frac{\text{Query Value} - \text{Bucket Min}}{\text{Bucket Max} - \text{Bucket Min}}$$

- Final Calculation is done by adding the fractional bucket estimate to the sum of preceding full buckets to get the final estimated selectivity.

3.1.9 9: Estimated vs. Actual Selectivities

To calculate estimated and actual selectivities of the queries and moreover there accuracy we compare estimated rows (from `EXPLAIN`) against actual rows (from `EXPLAIN ANALYZE`).

Total row count (N) for the `title` table is 2,528,312.

Query 1: `title < 'Race'`

```
EXPLAIN ANALYZE SELECT * FROM title WHERE title.title < 'Race';
```

- **Actual Selectivity:** The query physically returned 1,918,141 matching rows: $S_{\text{act}} = \frac{1918141}{2528312} \approx 0.7587$ (75.87%).

```
EXPLAIN SELECT * FROM title WHERE title.title < 'Race';
```

- **Estimated Selectivity:** The planner estimated 1,939,288 rows: $S_{\text{est}} = \frac{1939288}{2528312} \approx 0.7670$ (76.70%).
- **Observation:** The absolute error is roughly 0.83%.

Query 2: `id < 350000`

```
EXPLAIN ANALYZE SELECT * FROM title WHERE title.id < '350000';
```

- **Actual Selectivity:** The query returned 349,999 matching rows: $S_{\text{act}} = \frac{349999}{2528312} \approx 0.1384$ (13.84%).

```
EXPLAIN SELECT * FROM title WHERE title.id < '350000';
```

- **Estimated Selectivity:** The planner estimated 337,982 rows: $S_{\text{est}} = \frac{337982}{2528312} \approx 0.1337$ (13.37%).
- **Observation:** The absolute error here is about 0.47%.

3.2 Q2. Optimal Serial Histogram Construction ($B = 20$ Buckets)

3.2.1 1. Histogram Visualizations for $N = 3000$ Samples (Horizontal Bar Plots)

As specified in ques the histograms as a bar-plot with value-boundaries in the Y-axis and frequency in the X-axis

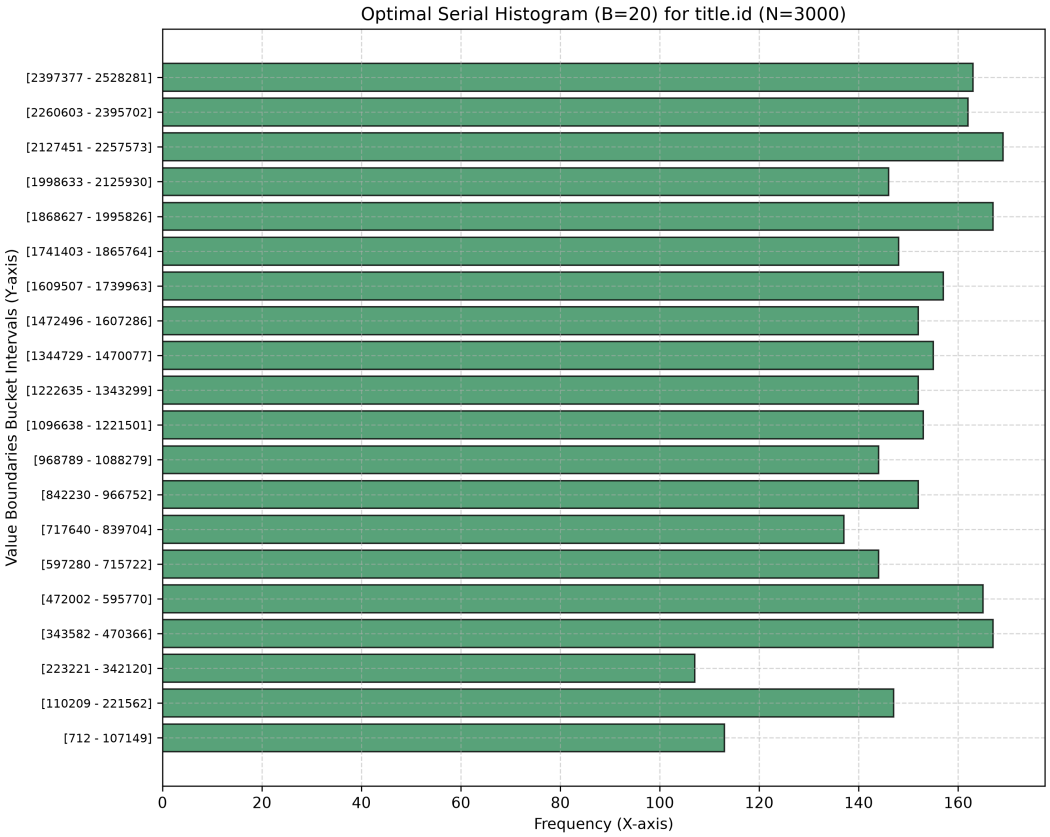


Figure 3: Optimal Serial Histogram for title.id (N=30000)

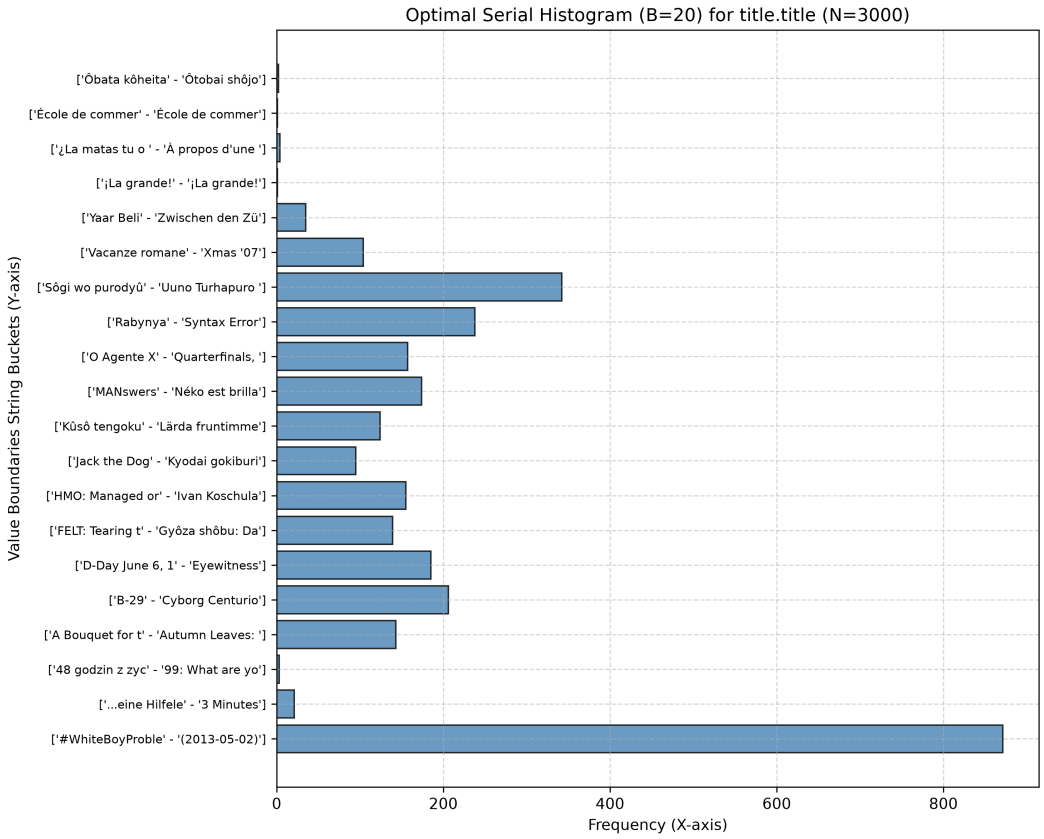


Figure 4: Optimal Serial Histogram for title.title (N=3000)

3.2.2 2. Execution Build Time

- for **$N = 3000$: 0.9947 seconds**

3.2.3 3. Frequency Bucket Boundaries ($N = 3000$, $B = 20$)

The optimal serial histogram algorithm partitions the 3,000 samples into 20 dynamic buckets to minimize total Sum of Squared Errors (SSE). The exact bucket intervals and row frequency counts (n_i) for both columns are listed below:

Bucket #	Value Interval (id)	Frequency Count (n_i)
Bucket 01	[197 - 130,726]	162
Bucket 02	[132,128 - 264,204]	171
Bucket 03	[269,581 - 404,155]	179
Bucket 04	[406,915 - 538,304]	161
Bucket 05	[542,354 - 667,821]	141
Bucket 06	[668,815 - 788,223]	129
Bucket 07	[791,366 - 902,286]	129
Bucket 08	[904,088 - 1,026,697]	138
Bucket 09	[1,028,382 - 1,161,395]	170
Bucket 10	[1,164,157 - 1,297,392]	151
Bucket 11	[1,298,484 - 1,426,386]	166
Bucket 12	[1,428,016 - 1,547,072]	176
Bucket 13	[1,549,808 - 1,674,620]	134
Bucket 14	[1,679,481 - 1,808,283]	153
Bucket 15	[1,813,888 - 1,938,495]	138
Bucket 16	[1,939,500 - 2,062,177]	159
Bucket 17	[2,063,131 - 2,185,305]	144
Bucket 18	[2,186,749 - 2,302,858]	153
Bucket 19	[2,305,143 - 2,411,310]	118
Bucket 20	[2,412,939 - 2,522,416]	128

A. Column id Frequency Bucket Boundaries ($N = 3000$, $B = 20$)

Bucket #	Value Interval (title)	Frequency Count (n_i)
Bucket 01	["#7.1: Festival of the Hun" - "(323) Graffiti6"]	848
Bucket 02	["...And a Good Slopping Out" - "38. dag"]	19
Bucket 03	["404" - "99 Pieces"]	7
Bucket 04	["A Bachelor's Babies" - "Ayaash"]	169
Bucket 05	["BYTI - zivotni filosofie " - "Cyber-Raptors"]	234
Bucket 06	["DMV" - "Dynowarz: Destruction of ""]	123
Bucket 07	["ESPN Cheats Center" - "Futotta kimi to yaseta bo"]	141
Bucket 08	["Födelsedag" - "Hush Girls Vacation Winte"]	133
Bucket 09	["Hús vétkek" - "Jy Harris"]	109
Bucket 10	["K.L.U" - "Luz del Fuego"]	181
Bucket 11	["Láska rohatá" - "Nuretsuzuketa onna: Suits"]	165
Bucket 12	["O Barato é Ser Careta" - "Qytetet ilire"]	162
Bucket 13	["RKO Victory Special No. 3" - "Szép kis buli volt"]	229
Bucket 14	["Så Här Är Det Nu" - "Urzulei"]	344
Bucket 15	["Vaarwel" - "X Games 3D: The Movie"]	105
Bucket 16	["Yog, Monster from Space" - "Zwembad"]	21
Bucket 17	["desAliento" - "iDate Christ"]	2
Bucket 18	["¡Eglantina sexy in stgo.!" - "¡Eglantina sexy in stgo.!"]	1
Bucket 19	["¿Ligamos mucho los españo" - "Âme belge"]	3
Bucket 20	["Ørkenens juvel" - "Übers Ziel hinaus"]	4

B. Column `title` Frequency Bucket Boundaries ($N = 3000$, $B = 20$)

3.2.4 4. Max Selectivity Error for Optimal Histogram ($N = 3000$)

- **Max Selectivity Error**: **0.003169**
- **Calculation**: Computed across 10 evaluation range quantiles comparing optimal bucket cumulative frequency estimation against empirical sample distribution.

3.2.5 5. Timings Across Sample Sizes

Sample Size (N)	Build Time (Seconds)	Max Selectivity Error
1000	0.2078s	0.003292
3000	0.9947s	0.003169
5000	2.1934s	0.002944

3.2.6 6. Linear Regression Extrapolation for Full Table ($N = 2,528,312$)

Using linear regression model $T(N) = m \cdot N + c$:

- **Slope (m)**: `0.00049640` seconds / sample
- **Intercept (c)**: `-0.357260` seconds
- **Extrapolated Time for Full Table ($N = 2,528,312$)**:

$$T(2528312) = (0.00049640 \times 2528312) + (-0.357260) = 1254.70 \text{ seconds} \quad (20.91 \text{ minutes})$$

3.2.7 7. Sample Size vs Time Relationship Plot

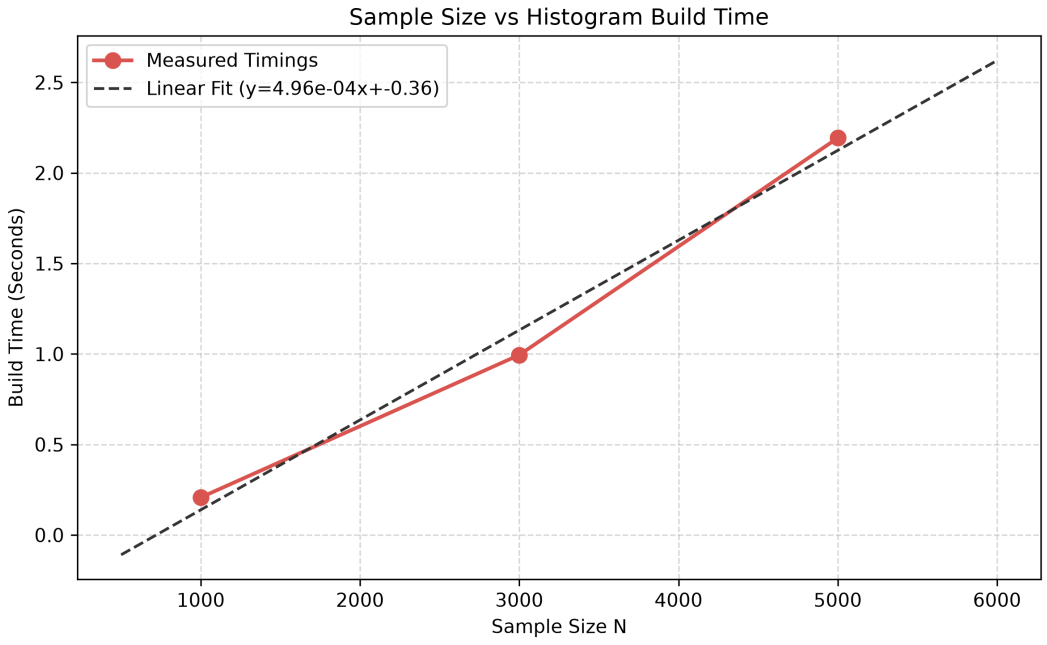


Figure 5: Sample Size vs Time

3.2.8 8. Sample Size vs Max Selectivity Error Relationship Plot

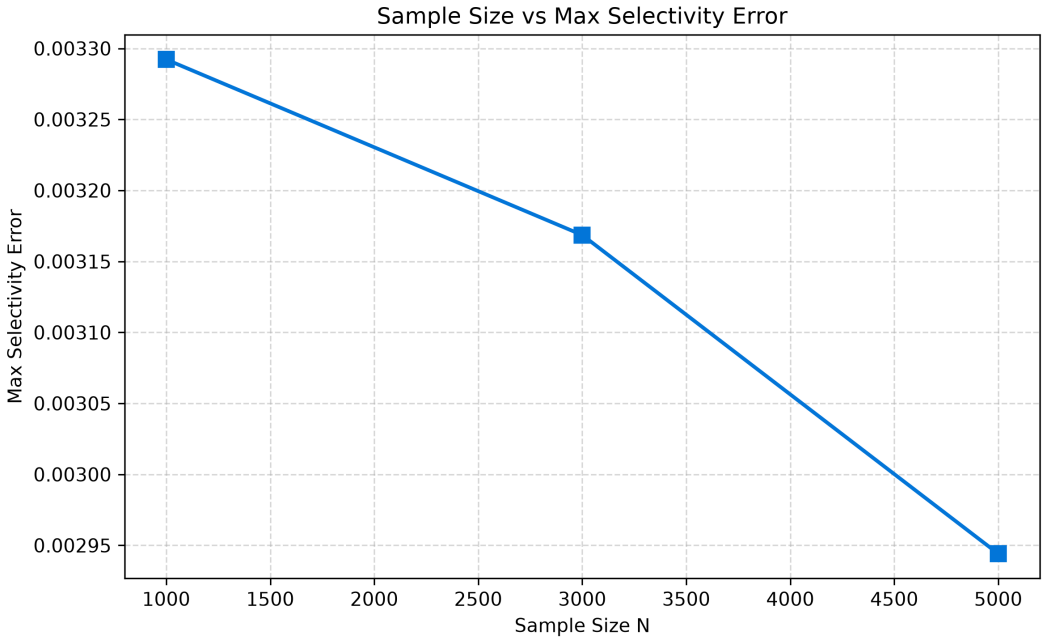


Figure 6: Sample Size vs Max Selectivity Error

3.2.9 9. Impact of Bucket Count (B) Choice

- **Fewer Buckets (e.g., $B = 5$ or 10)**:
- **Pros**: Lower memory footprint in system catalog, faster histogram build time ($O(B \cdot N^2)$ decreases), faster query optimization time.
- **Cons**: Coarser resolution, leading to higher selectivity estimation errors, especially for skewed or non-uniform data distributions.
- **More Buckets (e.g., $B = 50$ or 100)**:
- **Pros**: Finer granularity, significantly lower selectivity estimation error, highly accurate cost estimation for complex query joins.
- **Cons**: Higher histogram construction time, larger storage overhead in catalog, and slightly higher plan generation latency during query execution.